# Problem set 3

*Exponential maps and robot kinematics*

Modern Robotics

2013

**Before you start, please note:** while the numbers in these exercises were chosen such that you can complete the exercises by hand, it might be less work if you use the set of MATLAB scripts that was made available. Check the file `examples.m` for how to work with those functions—they could come in really handy in especially questions 3 and 4.[1]

## 1. Matrix exponentials

In Brockett's exponential formula (see question 3) we make use of the tilde form transformation of twists and the property of the matrix exponential given in (1). Verify that we may use this by proving the property.

$$e^{HAH^{-1}} = He^{A}H^{-1} \tag{1}$$

*Hint: start with the definition of the matrix exponential.*

Simply fill in $HAH^{-1}$ in the definition of the matrix exponential:

$$\begin{aligned}
e^{M} &:= I + M + \frac{M^2}{2!} + \frac{M^3}{3!} + \dots \\
e^{HAH^{-1}} &= I + HAH^{-1} + \frac{1}{2!}(HAH^{-1})(HAH^{-1}) + \dots \\
&= HIH^{-1} + HAH^{-1} + \frac{1}{2!}(HA)(H^{-1}H)(AH^{-1}) + \dots \\
&= H(I + A + \frac{1}{2!}A^2 + \dots)H^{-1} \\
&= He^{A}H^{-1} \quad \square
\end{aligned}$$

## 2. Rodriguez' formula

Rotation matrices can be quite difficult to calculate, especially when trying to remember which sine—or sign—goes where. However, in last homework set we saw that the time derivative of a rotation matrix[2] can be easily calculated.

[1] Although Rodriguez' formula is also in the functions, try to do question 2 "by hand"—you can use MATLAB, but don't use the supplied function.

[2] Alright, the question was on a homogeneous matrix, but the result holds if you substitute $H$ with $R$ and $\tilde{\omega}$ for $\tilde{T}$.

The resulting differential equation has a solution as in (2), where the exponential $e^{\tilde{\omega}}$ can be readily calculated with Rodriguez' formula as given in the slides.

$$R_{\mathrm{i}}^{\mathrm{j}}(t) = e^{\tilde{\omega}_{\mathrm{i}}^{\mathrm{j},\mathrm{j}} t} \tag{2}$$

Suppose that we execute a rotation by applying a rotational velocity of $\omega_{\mathrm{i}}^{\mathrm{j},\mathrm{j}} = (\frac{\pi}{6}, 0, 0)^\top \mathrm{rad\,s^{-1}}$ for $2\,\mathrm{s}$; what would be the final rotation matrix $R_{\mathrm{i}}^{\mathrm{j}}$?

*Hint: To use Rodriguez' formula, you'll need to express* $\tilde{\omega}$ *as a unit rotation* $\hat{\omega}$ *with* $\|\omega\| = 1$*—choose* $\theta$ *such that* $\tilde{\hat{\omega}}\theta = \tilde{\omega}t$.

Step one: normalise $\omega$: $\hat{\omega} = \frac{\omega}{\|\omega\|}$; and $\theta = 2\|\omega\|$, obviously:

$$\hat{\omega}_{\mathrm{i}}^{\mathrm{j},\mathrm{j}} = (1, 0, 0)^\top \quad , \quad \theta = \frac{\pi}{3} \tag{3}$$

Now we can directly apply Rodriguez' formula:

$$\begin{aligned} e^{\tilde{\omega}t} &= e^{\tilde{\hat{\omega}}\theta} \\ &= I_3 + (1, \tilde{0}, 0)^\top \sin\theta + {(1, \tilde{0}, 0)^\top}^2 (1 - \cos\theta) \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2}\sqrt{3} \\ 0 & \frac{1}{2}\sqrt{3} & \frac{1}{2} \end{pmatrix} \end{aligned} \tag{4}$$

Which indeed looks like a rotation around $\hat{x}$ of in total $60°$ or $\frac{\pi}{3}$ rad.

## 3. Brockett's exponential formula

Consider the crane depicted in Figure 1, which is a type often seen in for example harbours: it consists of a base that can ride on a track ($q_1$ [m]) and a pillar of fixed height with a rotating boom head ($q_2$ [rad]). The cable hangs from a pulley that can move along the boom ($q_3$ [m]) and of course the hook can be lowered ($q_4$ [m]).

Use Brockett's exponential formula to express the orientation (position+rotation) of $\Psi_4$, attached to the hook, with respect to $\Psi_0$, the inertial frame attached to the start of the track, when $q = (q_1, q_2, q_3, q_4)^\top = (3\,\mathrm{m}, -\frac{\pi}{2}\,\mathrm{rad}, 2\,\mathrm{m}, 3\,\mathrm{m})^\top$.

*Hint: First find the unit twists* $\tilde{\hat{T}}_i^{0,i\text{-}1}$ *and then calculate each matrix exponential as explained in the slides.*

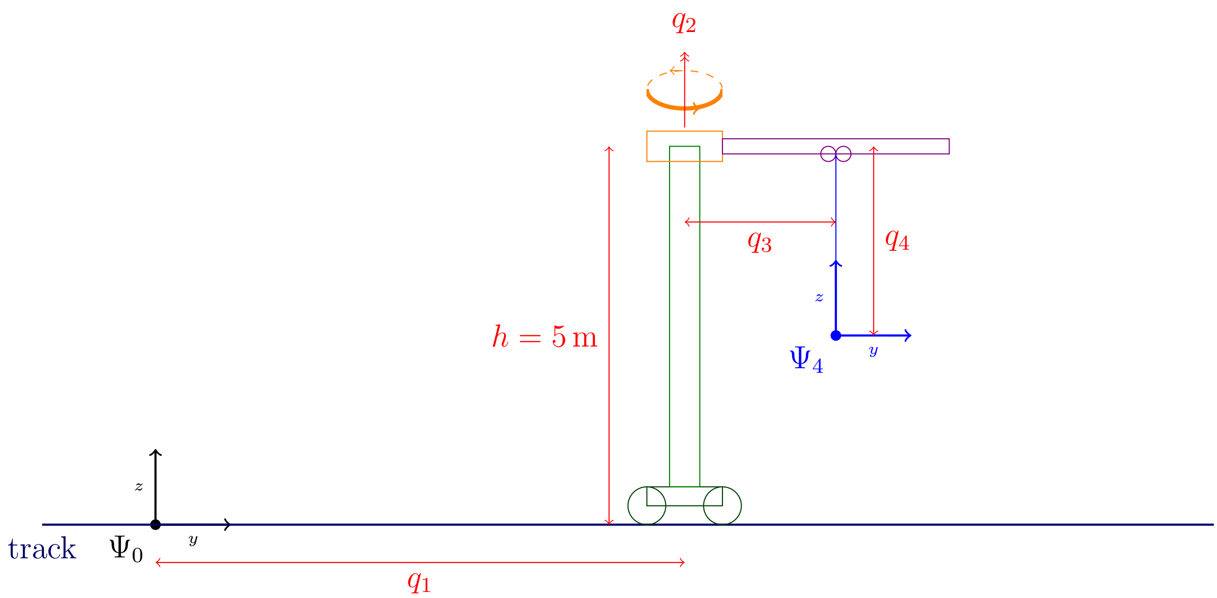


Figure 1: The crane considered in questions 3 and 4. Note that the joint positions $q_i$ are drawn arbitrarily—with the exception of $q_2$, which is 0 rad as drawn and hence both frames' $\hat{x}$ axes point out of the paper right now.

The most important part here is to find the unit twists $\tilde{\hat{T}}_{\mathrm{i}}^{0,\mathrm{i\text{-}1}}$, which are the joints' unit twists expressed in the inertial frame *in the reference configuration* for $q_i = 0$, $\forall i$. This last thing, the fact that we express the unit twists in the reference configuration, makes it really easy:

$$\hat{T}_1^{0,0} = (0,0,0,0,1,0)^\top \qquad \textit{translation along } +y \tag{5}$$

$$\hat{T}_2^{0,1} = (0,0,1,0,0,0)^\top \qquad \textit{rotation around } +z \tag{6}$$

$$\hat{T}_3^{0,2} = (0,0,0,0,1,0)^\top \qquad \textit{another translation along } +y \tag{7}$$

$$\hat{T}_4^{0,3} = (0,0,0,0,0,-1)^\top \qquad \textit{translation along } -z \tag{8}$$

Now all you need to fill in Brockett's formula is the initial configuration $H_4^0(0)$, which is easy to find by considering Figure 1:

$$H_4^0(0) = \begin{pmatrix} I_3 & \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix} \\ 0 & 1 \end{pmatrix} \tag{9}$$

Now if you calculate the matrix exponentials in the formula (using the supplied function `expT` or MATLABS's `expm` function[3]) with the tilde form of the unit twists in (5–8) and joint positions as given in the question, you should find for $H_4^0(q)$:

$$H_4^0(q) = \begin{pmatrix} 0 & 1 & 0 & 2 \\ -1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{10}$$

***Note****: Because of a mistake in our original solution to this question, we gave the misleading hint to include a frame $\Psi_{1b}$ and insert $H_{1b}^1$ in Brockett's formula. This is wrong, because that fixed height is already present in $H_4^0(0)$. What you could still do is treat the fixed pillar as a joint with fixed $q_{1b} = 5\,\mathrm{m}$, so you would add an extra $e^{\tilde{T}_{1b}^{0,1} \cdot 5}$ in the formula—but then the reference configuration $H_4^0(0)$ would be for $q_{1b} = 0$, i.e. turn out to be the identity.*

## 4. Geometric Jacobian

As a crane operator it is very important to know the speed of the load (hook), because if it moves too fast, it might bump into something and cause large damage. Of course the operator only has speed control over the 4 joints, so he needs your help to calculate the end-effector twist $T_4^{0,0}$.

Given the configuration in question 3, what is the instantaneous end-effector twist $T_4^{0,0}$ when the joint velocities are $\dot{q} = (2\,\mathrm{m\,s^{-1}}, -1\,\mathrm{rad\,s^{-1}}, 0\,\mathrm{m\,s^{-1}}, 1\,\mathrm{m\,s^{-1}})^\top$?

*Hint: First construct the Geometric Jacobian $J(q)$, of which $T_2$ and $T_3$ depend on the configuration $q$.*

Note that the in result $T_4^{0,0} = (\omega, v)^\top$ the velocity $v$ is defined in a special, geometric way rather than using $\dot{r}$ (see page 33 of Les2.pdf), so don't be confused by numbers in the outcome that may seem strange—try transforming the twist to $\Psi_4$ for a more intuitive representation.[4]

For the geometric Jacobian you will need to find $T_1 \ldots T_4$, which are—more than just the columns of $J(q)$—the unit twists of each joint expressed in the inertial frame. **Keep in mind** that, unlike in Brockett's exponential formula—where the coordinate transformation from $i$ to 0 was due to $H_i^0(0)$—in this case the Adjoints will depend on

[3]Which would be a waste of computing power, of course, because it does not make use of the special properties of $\tilde{\tilde{T}}$.

[4]Using $\mathrm{Ad}_{H_0^4}$ of course.

the configuration, because $H_\mathrm{i}^0$ depends on configuration. In other words: while $\hat{T}_\mathrm{i}^{\mathrm{i\text{-}1,i\text{-}1}}$ is constant, $T_i := \hat{T}_\mathrm{i}^{0,\mathrm{i\text{-}1}} = \mathrm{Ad}_{H^0_{\mathrm{i\text{-}1}}} \hat{T}_\mathrm{i}^{\mathrm{i\text{-}1,i\text{-}1}}$ is not, because $H_\mathrm{i}^0$ is a function of $q$!

With that in mind, the unit twists are still relatively easy to find:

$$
\begin{aligned}
T_1 &= (0,0,0,0,1,0)^\top && (11)\\
T_2 &= (\omega_2, (0,q_1,0)^\top \wedge \omega_2) \qquad\qquad \omega_2 = (0,0,1)^\top && (12)\\
T_3 &= (0,0,0,-\sin q_2, \cos q_2, 0)^\top && (13)\\
T_4 &= (0,0,0,0,0,-1)^\top && (14)
\end{aligned}
$$

You notice that due to the rotation of the boom head the direction of $T_3$ (expressed in inertial frame $\Psi_0$) changes; and that the proper expression for $T_2$ includes not only a rotation, but also a velocity component arising from $r \wedge \omega$.[5]

If you fill in the current configuration from Q3 and joint velocities as given in Q4, you should find that $T_4^{0,0} = (\omega, v)^\top = (0,0,-1,-3,2,-0.5)^\top$.

Now, if you consider the crane and the joint velocities, you might find that with the combination of $\dot{q}_1 = 2\,\mathrm{m\,s^{-1}}$ and $\dot{q}_2 = -1\,\mathrm{rad\,s^{-1}}$ in the current configuration, the hook should not move in $\hat{x}$ and $\hat{y}$ at all, so why isn't $T_4^{0,0} = (\bullet,\bullet,\bullet,0,0,\bullet)$?

This is again due to the fact that $v = r \wedge \omega + \lambda\omega$ and not simply $\dot{r}$: it is the velocity of a point *passing through the origin of* $\Psi_0$, but fixed to $\Psi_4$. In other words: due to the rotation of $\Psi_4$, it has a nonzero $v_x$ and $v_y$.

Expressing the velocity in $\Psi_4$ where $r = 0$ will give you $T_4^{4,0} = \mathrm{Ad}_{H_0^4} T_4^{0,0} = (0,0,-1,0,0,-0.5)^\top$: indeed the hook is (instantaneously) standing still in the lateral plane and only executing a rotation around $\hat{z}$ and a translation along $\hat{z}$—it is a downward-pointing screw around $\Psi_4$'s z-axis with $\omega = (0,0,-1)^\top$ and $\lambda = \frac{1}{2}$.

The solutions to questions 2–4 are also presented in a MATLAB file `P3.m`, using the supplied "toolbox" of geometric functions.

[5] Here you see why the velocity component of a twist should be $r \wedge \omega + \lambda\omega$ rather than $\dot{r}$: the effect of $\dot{q}_2$ on the end-effector definitely depends on $q_1$.